



Data Mining

HW1: Heart Disease Data Integration and KNN Classification

Supervisor: Dr. Fakhrahmad

Deadline: Thursday, 1404/08/22

1. Introduction

Heart disease continues to be one of the most significant global health concerns, accounting for a large proportion of mortality worldwide. Advances in data-driven methods have enabled researchers to analyze medical datasets more effectively, uncovering meaningful patterns that can assist in early detection and prevention. This project applies data mining techniques, specifically the K-Nearest Neighbors (KNN) classification algorithm, to the Heart Disease Dataset from the UCI Machine Learning Repository. The study aims to explore how patient characteristics such as age, sex, cholesterol, and blood pressure, can be used to predict the likelihood of heart disease. Through careful data preprocessing, exploratory analysis, and evaluation of model performance, the project demonstrates how machine learning can contribute to improving clinical decision support and understanding of cardiovascular risk factors.

2. Dataset Description

The dataset used in this study is the Heart Disease Dataset from the UCI Machine Learning Repository. It consists of four subsets collected from different medical centers: Cleveland, Hungarian, Switzerland, and VA Long Beach. Each subset includes similar clinical attributes that describe patient health conditions such as age, sex, cholesterol level, resting blood pressure, and other related features.

These four subsets are combined into a single dataset containing approximately 1000 records, with 13 main attributes and a target variable.

For more detailed information about the dataset's structure, features, and source, refer to the UCI Machine Learning Repository. This information should also be included in the report to properly acknowledge the dataset.

Dataset download link : [Link](#)

3. Data Preprocessing

Data preprocessing in this project involves preparing the dataset for analysis by ensuring data quality and consistency. It focuses on loading the data, handling missing or noisy values, and normalizing numerical features to improve model performance.

The main steps are as follows:

- **Loading the data:**
The dataset can be loaded using the Pandas library in Python (e.g., `pd.read_csv()`), which provides efficient tools for reading and managing tabular data.
- **Handling missing and noisy values:**
After loading, the dataset should be checked for missing values and noisy entries. If missing values are detected, they can be filled using imputation methods such as mean, median, mode imputation, or K-Nearest Neighbors (KNN) imputation. Any noisy or inconsistent records that could negatively impact the analysis should be identified and removed.
- **Normalizing numerical features:**
Normalization is applied to scale the numerical data and enhance model performance, especially for distance-based algorithms like KNN. Common normalization methods include Min-Max Scaling, Z-score Standardization, and Robust Scaling.

Each preprocessing step should be clearly documented in the report, explaining what was done, why it was necessary, and how it affects the data quality and model performance.

4. Exploratory Data Analysis (EDA)

Before applying the KNN algorithm, several exploratory data analysis (EDA) visualizations should be used to better understand the dataset and its characteristics. The following plots are recommended for initial exploration:

- **Histogram:** Useful for visualizing the distribution of numerical features such as age and cholesterol, helping to identify skewness or irregular patterns.
- **Boxplot:** Helps in detecting potential outliers and understanding the spread of continuous variables.
- **Q-Q Plot:** Assists in checking whether the data follows a normal distribution, which is important for some statistical analyses.
- **Correlation Matrix:** Displays relationships among numerical features, revealing possible multicollinearity or strong associations.

In addition to these, it is also recommended to consider other visualizations such as pair plots (scatterplot matrices) to observe relationships between multiple variables simultaneously, or count plots to summarize categorical attributes like sex or target class.

All visualizations should be generated using Python libraries such as Matplotlib or Seaborn, and included in the implementation section of the report. Each plot should be briefly explained to justify its purpose and to highlight which visualizations are most effective for describing this particular dataset.

5. Classification using K-Nearest Neighbors (KNN)

After integrating and preprocessing the datasets, the K-Nearest Neighbors (KNN) algorithm is applied for classification. This step aims to predict the presence or absence of heart disease based on the patients' clinical attributes.

The dataset includes a combination of numerical, binary, and categorical features. These attributes represent different kinds of medical indicators, and their scaling or encoding can influence the similarity measure between patients.

The general process involves the following steps:

- Splitting the dataset into training (80%) and testing (20%) sets. The training set is used to build the model, while the testing set evaluates its predictive performance.
- Choosing different values for K (e.g., 3, 5, 7) and comparing performance. Observe how the choice of K influences the classification outcomes.
- Assess the classifier's performance using appropriate metrics such as accuracy, precision, recall, and F1-score. These measures provide insight into how well the model distinguishes between patients with and without heart disease.
- Present the results visually by displaying a confusion matrix to illustrate the distribution of true and predicted classes. Additionally, plot the accuracy versus K-value to analyze how model performance changes with different neighborhood sizes.

KNN is a simple yet powerful classification algorithm based on the idea that instances with similar characteristics are likely to belong to the same class. In the context of medical data, such as this heart disease dataset, it identifies how patients with comparable clinical attributes tend to share similar diagnostic outcomes. However, the performance of KNN is highly sensitive to how similarity is measured. Because this dataset includes a mix of continuous and categorical or binary features, a single distance metric such as Euclidean distance may not capture all types of relationships accurately.

To handle this, different distance measures can be applied to different parts of the data. The results from these distances can then be combined using a weighted average, allowing each feature type to contribute appropriately to the overall similarity calculation. Report the optimal K value and describe the implementation steps.

6. Bonus Points (Optional)

To gain additional points and demonstrate a deeper understanding of the project, the following improvements can be implemented. A maximum of two of these enhancements will be considered for additional points:

- **Write Clean and Modular Code:**

Structure the implementation using well-organized, reusable functions and clear documentation to improve readability and maintainability.

- **Experiment with Different Preprocessing Techniques:**
Apply and compare multiple preprocessing strategies—such as various imputation or normalization methods—and analyze how they influence model performance.
- **Explore Additional Visualizations for EDA:**
Investigate other types of plots to gain deeper insights into feature relationships, distributions, and patterns within the dataset.
- **Use the Elbow Method to Determine the Best K:**
Apply the Elbow Method to identify the optimal value of K for the KNN classifier by analyzing how performance changes with different neighborhood sizes.
- **Apply K-Fold Cross-Validation:**
Use cross-validation to obtain more reliable performance estimates by evaluating the model across multiple training and testing splits.

Notes:

- Allowed programming language: **Python**
- Allowed libraries: NumPy, Pandas, Matplotlib, and Seaborn. If you wish to use any other libraries, please tell us for approval before doing so.
- Any sign of cheating will result in a zero grade for this assignment.
- **Submission:**
All submissions must be uploaded to Quera as a ZIP file named in the following format:
FirstNameFamilyName-StudentNumber.zip
Example: RezaZahedi-40432506.zip
The ZIP file should include both your implementation code and the project report.
- Your report should be in PDF format and include a summary of the implementation steps, final results, and answers to any assignment questions within their sections.

7. References

- [1] UCI Machine Learning Repository: Heart Disease Data Set.
- [2] J. Detrano et al., "International Application of a New Probability Algorithm for the Diagnosis of Coronary Artery Disease," American Journal of Cardiology, 1989.
- [3] Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques (3rd ed.). Burlington, MA: Morgan Kaufmann.